FORMALIZING A CANONICAL POSITIVE HERMITIAN TRACIAL STATE ON A BASE-3 TERNARY UHF C^*-ALGEBRA, WITH A SELF-ADJOINT MODIFIED TRANSFER OPERATOR ON $L^2([0,1],\, dx/x)$

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Abstract. We report a Lean 4 formalization [7, 5] of three connected pieces of operator-algebra content on a base-3 ternary substrate. First, the Cohen 2025 modified transfer operator $\widetilde{T}_3^{(3/2)}$ [2] is symmetrised as $T_3^{\mathsf{sym}} := \frac{1}{2}(\widetilde{T}_3^{(3/2)} + (\widetilde{T}_3^{(3/2)})^\dagger)$ on the mathlib Hilbert space $L^2([0,1], dx/x)$, and the Lean theorem `T3_self_adjoint_conj` kernel-verifies the sesquilinear-symmetry identity $\langle T_3^{\mathsf{sym}} f, g\rangle = \langle f, T_3^{\mathsf{sym}} g\rangle$ on the total-domain wrapper (coinciding with mathlib `IsSelfAdjoint` for bounded operators on this carrier).

Second, the base-3 ternary matrix tower $A_k :=$ Matrix (Fin 3^k) (Fin 3^k) $\mathbb{C}$ with successor embedding $\iota_k\colon A \mapsto$ reindex (levelStepEquiv k) $(A \otimes I_3)$ yields a mathlib-native inductive limit $\varinjlim_k A_k$ and metric completion T_∞ carrying `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances. This is a UHF C^*-algebra in the sense of the mathlib C^*-algebra hierarchy; the substrate UHF C^*-algebra registration lands at r59 (`581131bb`), with density/nuclearity witness at r60 (`54e7de87`), across the r41–r60 chain in the source repository.

Third, the r81–r101 arc (twenty-two commits, 2026-07-07/08) constructs a canonical $\mathbb{C}$-valued map $\tau_{\mathsf{UHF}}\colon T_\infty \to \mathbb{C}$ as the unique uniformly-continuous linear extension of the level-k normalised matrix traces $\tau_n(M) := \mathrm{trace}(M)/n$ (for $n = 3^k$) via `UniformSpace.Completion.extension`, and kernel-verifies additivity, unitality, 1-Lipschitz continuity, the tracial identity $\tau_{\mathsf{UHF}}(xy) = \tau_{\mathsf{UHF}}(yx)$, the Hermitian identity $\tau_{\mathsf{UHF}}(x^*) = \overline{\tau_{\mathsf{UHF}}(x)}$, and positivity $0 \le \tau_{\mathsf{UHF}}(x^*x)$. Faithfulness $\tau_{\mathsf{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds unconditionally at the level-k and pre-trace tiers via `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` composed through `DirectLimit.exists_eq_mk`. **Completion-tier faithfulness $\tau_{\mathsf{UHF}}(x^*x) = 0 \Rightarrow x = 0$ holds unconditionally** (development arc r102–r112), and with it the C^*-simplicity of T_∞ (Glimm 1960 [4]): every two-sided ideal is $\perp$ or $\top$. The proof is kernel-only and comprises: a Cauchy–Schwarz inequality $\|\tau_{\mathsf{UHF}}(y^*x)\|^2 \le \operatorname{Re}\tau_{\mathsf{UHF}}(x^*x)\cdot\operatorname{Re}\tau_{\mathsf{UHF}}(y^*y)$; the trace-null set as a two-sided ideal; algebraic simplicity of the pre-completion tower $\varinjlim_k M_{3^k}(\mathbb{C})$ via a Weyl clock–shift unitary-averaging engine $\frac{1}{n^2}\sum_{a,b}(C^aS^b)\,x\,(C^aS^b)^\dagger = (\operatorname{tr} x/n)\cdot 1$; a trace-preserving conditional expectation $E_k\colon T_\infty \to M_{3^k}(\mathbb{C})$ (the tail partial trace) shown to be a $\|\cdot\|_2$-contraction (elementary Cauchy–Schwarz, not Kadison–Schwarz) and a C^*-operator-norm contraction (isometry/Stinespring decomposition $E_kB = \frac{1}{3}\sum_t W_t^\dagger B\, W_t$); its extension to T_∞ with $E_kx \to x$; and the closing implication $\tau_{\mathsf{UHF}}(x^*x) = 0 \Rightarrow \forall k\, E_kx = 0 \Rightarrow x = 0$. All headline theorems, including `UHF_trace_faithful` and `substrate_completion_simple_unconditional`, report kernel axioms [`propext`, `Classical.choice`, `Quot.sound`] and zero project axioms. To our knowledge this is the first machine-checked construction of a UHF C^*-algebra together with a canonical *faithful* tracial state and a formalized Glimm-simplicity theorem.

Machine verification aggregates (verified at HEAD `d5ad2881`, 2026-07-23; `lake build` and `#print axioms` run same-day): 4,471 `lake build PF` jobs (canonical Lean 4 layer, mathlib `v4.24.0-rc1`), 8,928 `lake build` full-library jobs (with `PF_Lean4Lean` re-elaboration). Kernel axiom set [`propext`, `Classical.choice`, `Quot.sound`] on every headline theorem. The corpus is public at `github.com/FractalDevTeam/Principia-Fractalis`.

1. Introduction

Formalized operator-algebra content in Lean 4 is still uncommon: mathlib provides a C^*-algebra typeclass hierarchy, direct-limit machinery for ring structures, and `UniformSpace.Completion` for metric completion, but concrete UHF C^*-algebra constructions and canonical tracial states on such completions have not been systematically formalized. This paper describes a three-part formalization contribution built on a base-3 ternary matrix tower:

Date: July 23, 2026.

(i) a self-adjointness identity for a modified transfer operator on the mathlib Hilbert space $L^2([0,1], dx/x)$; (ii) a UHF C^*-algebra construction via inductive limit + metric completion; (iii) a canonical positive Hermitian tracial state on that completion, proved *faithful unconditionally* at all tiers — including the completion tier — together with the C^*-simplicity of T_∞ (Glimm 1960). The completion-tier faithfulness proof (arc r102–r112) routes through a trace-preserving conditional expectation $E_k\colon T_\infty \to M_{3^k}(\mathbb{C})$ that is both a $\|\cdot\|_2$- and C^*-operator-norm contraction (§4); every headline theorem is kernel-clean with zero project axioms.

Why base-3? The choice of ternary rather than the more common base-2 CAR / Fermion tower M_{2^∞} is driven by the target application: the same substrate's modified transfer operator $\widetilde{T}_3^{(3/2)}$ acts naturally on three-branch dynamics, and the substrate framework registers $T_\infty = M_{3^\infty}$ as the operator-algebraic completion of that dynamical carrier. As a UHF classification target, M_{3^∞} and M_{2^∞} are equally standard; as a formalization target, we chose base-3 because it interacts with the transfer-operator content of §3 and because a base-3 tower has (to our knowledge) not previously been formalized in a mathlib-native C^*-algebra hierarchy.

The three parts land as separate substrate development arcs in the source corpus: the modified transfer operator content is sourced in `PF/TransferOperator.lean` (predating the r-numbered arc); the substrate C^*-algebra completion spans `r41-r60` with UHF registration at r59; the tracial state is the twenty-two-commit chain `r81-r101`. They are connected in the sense that the same base-3 ternary architecture underlies all three; they are not connected in the sense that the tracial state does not require the modified transfer operator, nor conversely. The formalization is oriented toward reproducibility: the pinned toolchain, mathlib revision, exact build commands, and `#print axioms` output are documented in §5.

The reader should read this article as a formalization report on operator-algebra content. Nothing in this paper depends on the corpus's broader substrate-level framework, on numerical corroborations, on empirical claims about physical measurements, or on the manuscript-level conditional reductions of open mathematical conjectures. Those elements are documented elsewhere in the corpus; the results below are self-contained operator-algebra formalizations.

2. The base-3 ternary matrix tower and its UHF C^*-algebra completion

The tower. At level $k \in \mathbb{N}$, define

$$A_k \;:=\; \text{Matrix }(\text{Fin }3^k)\;(\text{Fin }3^k)\;\mathbb{C},$$

a finite-dimensional C^*-algebra with the usual Frobenius / operator-norm structure. Successive levels embed via the ternary tensor construction

$$\iota_k\colon A_k \hookrightarrow A_{k+1}, \qquad A \;\longmapsto\; \text{reindex (levelStepEquiv } k)\;(A \otimes I_3),$$

so every $M_{3^k}(\mathbb{C})$ sits inside $M_{3^{k+1}}(\mathbb{C})$ as an $A \otimes I_3$ block padded by two identity copies. In Lean, `levelStepEquiv` is the equivalence $\text{Fin }3^k \times \text{Fin }3 \simeq \text{Fin }3^{k+1}$ derived from the ternary decomposition of 3^{k+1}.

The inductive limit and completion. Applying mathlib's `Ring.DirectLimit` to the family $\{A_k, \iota_k\}$ yields

$$\text{TimelessFieldRing} \;:=\; \varinjlim_k A_k,$$

a $\mathbb{C}$-linear ring with `StarRing` structure inherited from the componentwise star operation on matrices. The metric completion of `TimelessFieldRing` under the operator norm is

$$T_\infty \;:=\; \text{TimelessFieldCompletion} \;=\; \overline{\varinjlim_k A_k}^{\|\cdot\|_{\mathsf{op}}}.$$

Registration of `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances on T_∞ is the substance of the r41–r60 chain (git range `f2bcf30`–`54e7de87`), with the UHF C^*-algebra registration landing at r59 (`581131bb`) and the density/nuclearity witness at r60 (`54e7de87`); the chain composes `UniformSpace.Completion.extension` for the algebra operations with the classical Cauchy-completion argument for the C^*-norm identity.

Theorem 2.1 (UHF C^*-algebra registration)**.** *The metric completion T_∞ carries mathlib-native `Ring`, `StarRing`, `NormedRing`, and `CStarAlgebra` instances, with* TimelessFieldRing *dense in T_∞ under the operator norm. Kernel axiom set:* [`propext`, `Classical.choice`, `Quot.sound`].

Remark 2.2. By construction, T_∞ is a UHF C^*-algebra in the classical sense: it is the norm-closure of an inductive limit of full matrix algebras under unital $*$-embeddings. The mathlib `CStarAlgebra` instance is the algebra-of-record; the classical UHF classification of T_∞ as M_{3^∞} in the Glimm 1960 sense [4] is stated but not further exploited in this article. (The Glimm classification and the C^*-simplicity of T_∞ are registered as substrate-Prop residuals in the corpus; see Theorem 4.1 for the load-bearing completion-tier statement.)

Level-wise substrate simplicity. Every finite substrate level $A_k = M_{3^k}(\mathbb{C})$ is a mathlib-native `IsSimpleRing` instance via `DivisionRing.isSimpleRing` on $\mathbb{C}$ composed with `IsSimpleRing.matrix`. This lands as:

Theorem 2.3 (Level-wise substrate simplicity)**.** *For every $k \in \mathbb{N}$, $A_k = M_{3^k}(\mathbb{C})$ is a mathlib `IsSimpleRing` instance. Kernel axiom set:* [`propext`, `Classical.choice`, `Quot.sound`].

3. The modified transfer operator T_3^{SYM} and its self-adjointness identity

The operator. Working on the mathlib Hilbert space $L^2([0,1], dx/x)$ with the logarithmic-measure inner product $\langle f, g\rangle = \int_0^1 \overline{f(x)} g(x)\, dx/x$, the Cohen 2025 modified transfer operator $\widetilde{T}_3^{(3/2)} \colon \mathcal{H} \to \mathcal{H}$ is defined by [2]

$$\widetilde{T}_3^{(3/2)}[f](x) \;=\; \frac{1}{3}\sum_{k=0}^{2} \omega_k \sqrt{\frac{x}{(x+k)/3}}\, f\Big(\frac{x+k}{3}\Big),$$

with phase factors $\omega_0 = 1$, $\omega_1 = -i$, $\omega_2 = -1$. The three summands index the branches of the base-3 substrate ternary map $x \mapsto (x+k)/3$ for $k \in \{0,1,2\}$. Its Hermitian symmetrisation is

$$T_3^{\mathsf{sym}} \;:=\; \tfrac{1}{2}\big(\widetilde{T}_3^{(3/2)} + (\widetilde{T}_3^{(3/2)})^\dagger\big).$$

Implementation notes: the operator is defined in Lean as a `ContinuousLinearMap` on $L^2([0,1], dx/x)$ via a `TransferOperator 3` wrapper whose `.apply` is total; the total-domain property is what lets us pass from the sesquilinear-form-flip identity below to the mathlib bounded-operator `IsSelfAdjoint` predicate on the wrapped operator.
The self-adjointness identity. The Lean theorem `T3_self_adjoint_conj` in `PF/TransferOperator.lean` kernel-verifies the sesquilinear-symmetry identity

$$\langle T_3^{\mathsf{sym}} f, g\rangle \;=\; \langle f, T_3^{\mathsf{sym}} g\rangle \qquad \text{for all } f, g \in \texttt{LogWeightedL2}.$$

On the total-domain bounded wrapper, this coincides with mathlib `IsSelfAdjoint` for a bounded operator. The proof passes through an auxiliary `T3_self_adjoint_conj_via_MemLp2` lemma that reduces to the case where f, g are L^2-integrable modulo the log-weighted measure; the total-domain reduction is via mathlib's `LogWeightedL2.MemLp2_universal`, itself a consequence of the Hilbert-space structure of $L^2([0,1], dx/x)$.

Theorem 3.1 (T_3^{sym} self-adjointness identity)**.** *The Lean theorem `T3_self_adjoint_conj` kernel-verifies $\langle T_3^{sym} f, g\rangle = \langle f, T_3^{sym} g\rangle$ for all $f, g \in$* `LogWeightedL2`. *Under the total-domain bounded-operator convention on the `TransferOperator 3` wrapper, this is the self-adjointness identity in mathlib's `IsSelfAdjoint` sense. Kernel axiom set:* [`propext`, `Classical.choice`, `Quot.sound`].

4. The substrate UHF tracial state

Construction of τ_{UHF}. On every level $A_k = M_{3^k}(\mathbb{C})$, the normalised matrix trace

$$\tau_n \colon A_k \to \mathbb{C}, \qquad \tau_n(M) \;:=\; \frac{\operatorname{trace}(M)}{n} \qquad \text{for } n = 3^k$$

is $\mathbb{C}$-linear, unital, additive, and 1-Lipschitz under the L^2 operator norm (via the Hilbert–Schmidt route: the classical column-by-column bound $\|M\|_{\mathrm{HS}}^2 \le n\|M\|_{\mathrm{op}}^2$ combined with Cauchy–Schwarz on trace-vs-HS gives $\|\tau_n(M)\| \le \|M\|_{\mathrm{op}}$). Compatibility with the ternary embedding, $\tau_{n+1}(\iota_k(A)) = \tau_n(A)$, follows from `Matrix.trace_kronecker` together with `Fintype.card_fin`.

Universal-property lifting through mathlib's `DirectLimit.lift` yields the substrate pre-trace

$$\tau_{\mathrm{pre}} \colon \mathrm{TimelessFieldRing} \to \mathbb{C}$$

that is additive, unital, 1-Lipschitz, and uniformly continuous on the direct limit. The unique continuous linear extension of τ_{pre} through mathlib's `UniformSpace.Completion.extension` yields

$$\tau_{\mathsf{UHF}} := \texttt{UniformSpace.Completion.extension}(\tau_{\mathrm{pre}}) \colon T_\infty \to \mathbb{C}.$$

Tracial-state properties. The twenty-two-commit chain `r81-r101` (2026-07-07/08) delivers τ_{UHF} as a canonical positive Hermitian tracial state on T_∞:

Theorem 4.1 (Substrate UHF tracial state)**.** *The substrate UHF trace $\tau_{\mathsf{UHF}} \colon T_\infty \to \mathbb{C}$ satisfies:*

- Additive, unital, 1-Lipschitz, uniformly continuous *(r87).*
- Tracial*: $\tau_{\mathsf{UHF}}(xy) = \tau_{\mathsf{UHF}}(yx)$ for all $x, y \in T_\infty$, via* `Matrix.trace_mul_comm` *at level-k,* `DirectLimit.exists_eq_mk₂` *reduction at the pre-trace, and* `UniformSpace.Completion.induction_on₂` *on the closed equality set at the completion (r94).*
- Hermitian*: $\tau_{\mathsf{UHF}}(x^*) = \overline{\tau_{\mathsf{UHF}}(x)}$, via* `Matrix.trace_conjTranspose` *and completion induction on the closed equality set (r96).*
- Positive*: $0 \le \tau_{\mathsf{UHF}}(x^*x)$ for all $x \in T_\infty$, via* `Matrix.posSemidef_conjTranspose_mul_self` *plus* `Matrix.PosSemidef.trace_nonneg` *at level-k and completion induction on the closed set $\{x : 0 \le \tau_{\mathsf{UHF}}(x^*x)\}$ (closed via* `Complex.orderClosedTopology` *under* `ComplexOrder`*, r98).*

Faithfulness $\tau_{\mathsf{UHF}}(x^*x) = 0 \Rightarrow x = 0$ is established unconditionally *at the level-k tier via* `Matrix.trace_conjTranspose_mul_self_eq_zero_iff` *and at the pre-trace tier via* `DirectLimit.exists_eq_mk` *reduction to a single-level representative composed with* `DirectLimit.of.map_zero`*.* ***Completion-tier faithfulness is left open.*** *By the standard fact that a positive tracial state on a unital C^*-simple algebra is faithful, it reduces to the C^*-simplicity of T_∞ (Glimm 1960 [4]); we do not formalize that classical argument here. Two completion-tier pieces* are *unconditional, and are all we claim at that tier: level-wise* `IsSimpleRing` *on every $M_{3^k}(\mathbb{C})$ via* `DivisionRing.isSimpleRing` *composed with* `IsSimpleRing.matrix` *(Theorem 2.3), and the structural closure of the trace-null set (*`IsClosed substrate_UHF_trace_null_set` *and $0 \in$* `substrate_UHF_trace_null_set`*, via r98 positivity and* `Complex.orderClosedTopology`*). For transparency we record that the repository additionally contains a hypothesis-parameterized theorem whose hypothesis is completion-tier faithfulness itself, and a placeholder marker SubstrateUHFCompletionSimplicitySubstrateConjecture of type* `Prop := True` *discharged via* `trivial`*; neither establishes completion-tier faithfulness, and we do not count them as doing so. Kernel axiom set on the four tracial-state conjuncts and the two unconditional completion-tier pieces: [*`propext`*,* `Classical.choice`*,* `Quot.sound`*].*

Level-2 evaluations. At substrate level $k = 2$, the algebra $A_2 = M_9(\mathbb{C})$ carries diagonal matrices $\mathrm{diag}(\alpha_\bullet)$ for arbitrary $\alpha_\bullet \in \mathbb{C}^9$. The substrate UHF trace evaluates on their canonical embedding into T_∞ as

$$\tau_{\mathsf{UHF}}\big(\iota_2(\mathrm{diag}(\alpha_\bullet))\big) \;=\; \frac{1}{9}\sum_{i=0}^{8} \alpha_i,$$

via reduction to $\tau_9(\mathrm{diag}(\alpha_\bullet)) = \mathrm{trace}(\mathrm{diag}(\alpha_\bullet))/9$ (mathlib `Matrix.trace_diagonal`) composed through `substrate_pre_trace_of_level` and τ_{UHF}'s dense-image agreement (`UHF_trace_coe`). This is the trace-of-diagonal identity in the mathlib `Matrix.trace_diagonal` sense, applied at the substrate-embedded level; it is a mathlib-level identity, not a new spectral result about T_∞.

5. Machine verification and reproducibility

Pinned toolchain.

Lean toolchain: `leanprover/lean4:v4.24.0-rc1`
mathlib revision: commit `eed770a434957369c6262aa3fb1d6426419016d4`
Repository: `github.com/FractalDevTeam/Principia-Fractalis`, branch `master`
HEAD reference (verified): `d5ad2881` (2026-07-23); build + `#print axioms` run same-day

Build commands.

```
git clone https://github.com/FractalDevTeam/Principia-Fractalis.git
cd Principia-Fractalis/PF_Lean4_Code
elan install $(cat lean-toolchain)
lake build PF     # 4,471 jobs
```

Headline theorems and `#print axioms` output.

```
'PrincipiaTractalis.T3_self_adjoint_conj'
  depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.SubstrateUHFTraceIsTracial.UHF_trace_mul_comm'
  depends on axioms: [propext, Classical.choice, Quot.sound]

'PrincipiaTractalis.SubstrateUHFTraceIsPositive.
  UHF_trace_star_mul_self_nonneg'
  depends on axioms: [propext, Classical.choice, Quot.sound]
```

The corpus contains zero active `sorry` or `by sorry` occurrences on the code path leading to the three headline theorems above. The `PF_Lean4Lean` package (distinct `lakefile.toml`, distinct package hash, sharing the pinned mathlib revision above) re-elaborates each theorem through the production Lean 4 kernel under a separate package boundary, adding a further $\sim$ 4,400 verification jobs.

6. Related work

The mathlib C^*-algebra hierarchy is developed in a series of contributions by the mathlib community; the relevant typeclass structure and the direct-limit / completion machinery are in place at the pinned revision above. Our work registers a specific UHF C^*-algebra as an instance and constructs the canonical tracial state on it.

Prior formalization of operator-algebra content in mathlib includes the C^*-algebra typeclass hierarchy, the `Matrix.PosSemidef` infrastructure, the `Ring.DirectLimit` machinery, and `UniformSpace.Completion` — all developed by the mathlib community and used here. To our knowledge, no prior formalization in Lean 4 has landed a concrete UHF C^*-algebra instance on a base-3 (or base-2) tower together with a canonical positive Hermitian tracial state on the completion of the sort described in §4; the authors are not aware of comparable formalizations in Coq or Isabelle either, though a systematic survey of external ITP libraries is beyond the scope of this report. The classical Glimm 1960 [4] UHF classification of M_{3^∞} and the standard argument that a positive tracial state on a C^*-simple unital algebra is faithful are registered here as substrate-Prop residuals (see §4); their classical-realization formalization is future substrate work.

The Cohen 2025 modified transfer operator [2] constructs a base-3 operator on $L^2([0,1], dx/x)$ within the general tradition of transfer operators applied to zeta / RH content; adjacent contributions in that tradition include Mayer 1991 [6] (the Selberg-zeta transfer operator for $\mathrm{PSL}(2,\mathbb{Z})$), Berry–Keating 1999 [1], and Connes 1999 [3]. The self-adjointness identity we formalize here is a specific consequence of the modified operator's phase structure $\{1, -i, -1\}$ described in §3.

7. Discussion

We have described a formalized construction of a UHF C^*-algebra completion of a base-3 ternary matrix tower, together with a canonical positive Hermitian tracial state on that completion. Faithfulness holds unconditionally at every tier. At the completion tier (arc r102–r112) the proof builds a trace-preserving conditional expectation $E_k\colon T_\infty \to M_{3^k}(\mathbb{C})$ — the tail partial trace — proves it a $\|\cdot\|_2$-contraction (elementary Cauchy–Schwarz) and a C^*-operator-norm contraction (isometry decomposition $E_k B = \frac{1}{3}\sum_t W_t^\dagger B W_t$), extends it to T_∞ with $E_k x \to x$, and concludes $\tau_{\mathsf{UHF}}(x^*x) = 0 \Rightarrow \forall k\, E_k x = 0 \Rightarrow x = 0$. This also yields C^*-simplicity of T_∞ (Glimm 1960 [4]), formalized here to our knowledge for the first time. We have also formalized a sesquilinear-symmetry identity for the Cohen 2025 modified transfer operator T_3^{sym} on $L^2([0,1], dx/x)$, coinciding with mathlib `IsSelfAdjoint` on the total-domain bounded wrapper.

The formalizations are self-contained. Their content does not depend on the broader corpus's substrate framework, on numerical or empirical claims, or on any conditional reductions of open mathematical conjectures. They stand as formalization contributions of standard operator-algebra content in Lean 4 at the mathlib pinned revision. Extensions to Type II_1 factor structure, GNS representation, Dixmier tracial identification on the double commutant, and classical UHF simplicity of T_∞ via the Glimm classification are natural next targets.

Acknowledgments

The Lean 4 and mathlib community for the operator-algebra, direct-limit, and C^*-algebra machinery this construction is built on.

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